The Number e

From Compound Interest to Calculus

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Abstract

The number $e \approx 2.71828$ is one of the central constants of mathematics, sitting alongside π as a quantity that appears unbidden across analysis, probability, and applied science. Unlike π, which has an obvious geometric origin, e can feel mysterious: where does it come from, and why is it "natural"? This article builds the answer in stages. We begin with a concrete financial question about compound interest, extract a limit from it, give that limit a numerical value, and then connect it to an infinite series. From there we arrive at the property that truly distinguishes e in calculus—that the exponential function e^x is its own derivative—and finally we give a clean, rigorous definition by way of the natural logarithm defined as an integral.

1 A Question About Money

Imagine you deposit \$1 in a bank account that pays 100% annual interest. After one year you have \$2. Simple enough.

Now suppose the bank compounds the interest twice a year. Instead of 100% once, you get 50% added halfway through the year, and then 50% on the new, larger balance at the end:

$$\left(1+\tfrac{1}{2}\right)\left(1+\tfrac{1}{2}\right) = \left(1+\tfrac{1}{2}\right)^2 = 2.25.$$

Compounding more often earns you more, because you start earning interest on interest sooner. With quarterly compounding ($n = 4$):

$$\left(1+\tfrac{1}{4}\right)^4 = 2.44140\ldots,$$

and with monthly compounding ($n = 12$):

$$\left(1+\tfrac{1}{12}\right)^{12} = 2.61303\ldots$$

The natural question is: if we compound more and more frequently—daily, hourly, every second, *continuously*—does the final balance grow without bound, or does it settle toward some limit? The pattern in the table below already suggests the answer.

Compounding periods n	Balance $\left(1+\frac{1}{n}\right)^n$
1	2.000000
2	2.250000
4	2.441406
12	2.613035
365	2.714567
8,760	2.718127
1,000,000	2.718280

The numbers are increasing, but they are not running off to infinity. They are crowding up against a ceiling a little above 2.718. That ceiling is the number e.

2 The Limit Definition

The financial story gives us our first definition of e.

Definition 2.1 (Limit definition)**.**

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n.$$

This raises an immediate worry: it is not obvious that the limit even exists. The base $1+\frac{1}{n}$ is shrinking toward 1, while the exponent n is growing toward infinity. A quantity slightly larger than 1 raised to a huge power could in principle do anything. We need to know that these two effects balance to produce a finite number.

Proposition 2.2. *The sequence $a_n = \left(1+\frac{1}{n}\right)^n$ is increasing and bounded above, hence it converges.*

Sketch. Expanding a_n with the binomial theorem gives

$$a_n = \sum_{k=0}^{n}\binom{n}{k}\frac{1}{n^k} = \sum_{k=0}^{n}\frac{1}{k!}\cdot\frac{n(n-1)\cdots(n-k+1)}{n^k} = \sum_{k=0}^{n}\frac{1}{k!}\prod_{j=0}^{k-1}\left(1-\frac{j}{n}\right).$$

Each factor $\left(1-\frac{j}{n}\right)$ lies in $(0,1]$ and *increases* toward 1 as n grows; moreover increasing n adds an extra nonnegative term to the sum. Both effects push a_n upward, so the sequence is increasing. For an upper bound, drop the products (each is at most 1) and bound the factorials below by powers of two:

$$a_n \le \sum_{k=0}^{n}\frac{1}{k!} \le 1 + \sum_{k=1}^{n}\frac{1}{2^{k-1}} < 1+2 = 3.$$

An increasing sequence bounded above converges, by the monotone convergence theorem for real sequences. □

So the limit exists and lies between 2 and 3. "Continuous compounding" is not just a vivid phrase; it is a genuine mathematical limit, and its value is e.

3 The Series Definition

The proof above quietly handed us a second, often more convenient, definition. The expansion of a_n involved the partial sums of $\sum 1/k!$, and as $n\to\infty$ each product $\prod\left(1-\frac{j}{n}\right)$ tends to 1. Taking the limit term by term turns the limit definition into an infinite series.

Definition 3.1 (Series definition)**.**

$$e = \sum_{k=0}^{\infty}\frac{1}{k!} = \frac{1}{0!}+\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\cdots = 1+1+\frac{1}{2}+\frac{1}{6}+\frac{1}{24}+\cdots$$

This series converges very fast, because factorials grow explosively. Just the first few terms already pin down several digits:

$$1+1+0.5+0.1\overline{6}+0.041\overline{6}+0.008\overline{3}+\cdots = 2.71805\ldots,$$

and only a handful more terms give

$$e = 2.718281828459045\ldots$$

The two definitions describe the same number, but they are not interchangeable in difficulty. The limit definition is where e comes *from* (the modeling question), while the series is how you actually *compute* it. The equivalence of the two is the content of the proposition above made fully rigorous.

4 Irrationality

It is worth pausing to note that e, like π, is irrational: it cannot be written as a ratio of integers, and its decimal expansion never terminates or repeats. The series makes this surprisingly accessible.

Theorem 4.1. *e is irrational.*

Sketch. Suppose for contradiction that $e = p/q$ with positive integers p, q. Consider

$$x = q!\left(e - \sum_{k=0}^{q} \frac{1}{k!}\right).$$

If $e = p/q$ then $q!\,e = p\,(q-1)!$ is an integer, and $q!\sum_{k=0}^{q}\frac{1}{k!}$ is an integer as well (every denominator $k!$ with $k \le q$ divides $q!$). So x would be an integer. But writing out the leftover tail,

$$x = \sum_{k=q+1}^{\infty} \frac{q!}{k!} = \frac{1}{q+1} + \frac{1}{(q+1)(q+2)} + \cdots < \sum_{m=1}^{\infty} \frac{1}{(q+1)^m} = \frac{1}{q},$$

so $0 < x < 1$. There is no integer strictly between 0 and 1, a contradiction. Hence e is irrational. □

5 What Makes e "Natural": Calculus

So far e is a specific number that happens to arise from compounding. The reason mathematicians single it out, rather than 2.718 or any nearby value, only becomes clear in calculus.

Consider the family of exponential functions $f(x) = b^x$ for a base $b > 0$. Every one of them has a derivative proportional to itself:

$$\frac{d}{dx}\,b^x = \lim_{h\to 0}\frac{b^{x+h} - b^x}{h} = b^x \lim_{h\to 0}\frac{b^h - 1}{h} = C_b\, b^x,$$

where the constant $C_b = \lim_{h\to 0}\frac{b^h-1}{h}$ depends only on the base. For $b = 2$ one finds $C_2 \approx 0.693$, and for $b = 3$ one finds $C_3 \approx 1.099$. Somewhere between 2 and 3 there must be a special base for which this constant is exactly 1. That base is e, and it is the choice that makes the derivative formula as clean as possible.

Theorem 5.1 (The defining calculus property)**.** *The exponential function e^x is its own derivative:*

$$\frac{d}{dx}\,e^x = e^x.$$

Equivalently, e is the unique base b for which $\displaystyle\lim_{h\to 0}\frac{b^h - 1}{h} = 1.$

This is why e deserves the name "natural." Among all exponential functions, e^x is the one whose rate of change at every point equals its current value—no stray constant attached. Any process whose growth rate is proportional to its present size (radioactive decay, population models, charging capacitors, and yes, continuously compounded interest) is governed by e^x, because the equation $y' = ky$ has solution $y = y_0 e^{kx}$.

We can also see this property directly from the series. Differentiating

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

term by term shifts every term down by one,

$$\frac{d}{dx}\, e^x = 0 + 1 + x + \frac{x^2}{2!} + \cdots = e^x,$$

recovering the same function. (Setting $x = 1$ in the series gives back $\sum 1/k!$, our earlier value of e.)

6 A Rigorous Definition via the Natural Logarithm

The definitions above are honest, but each leans on something delicate—existence of a limit, convergence of a series, or term-by-term differentiation. A modern analysis course usually prefers a cleaner foundation that builds everything from a single integral, and recovers e at the end as a derived quantity. The starting point is the natural logarithm.

Definition 6.1 (Natural logarithm)**.** For $x > 0$, define

$$\ln x = \int_1^x \frac{1}{t}\, dt.$$

This is well posed because $1/t$ is continuous on $(0, \infty)$, so the integral exists for every $x > 0$. The definition has immediate, easily proved consequences. By the Fundamental Theorem of Calculus,

$$\frac{d}{dx}\, \ln x = \frac{1}{x} > 0,$$

so ln is strictly increasing, hence one-to-one. A change of variables $t \mapsto st$ in the integral yields the familiar law $\ln(xy) = \ln x + \ln y$, and from it $\ln(x^r) = r \ln x$. Because ln is strictly increasing and unbounded in both directions (the integral diverges as $x \to \infty$ and as $x \to 0^+$), it maps $(0, \infty)$ bijectively onto all of $\mathbb{R}$. It therefore has an inverse function, which we call exp.

Now e has a strikingly simple definition: it is the input at which the area under $1/t$ first reaches exactly 1.

Definition 6.2 (e via area)**.** e is the unique real number satisfying

$$\ln e = \int_1^e \frac{1}{t}\, dt = 1.$$

Such a number exists and is unique because ln is a continuous strictly increasing bijection onto $\mathbb{R}$, so the equation $\ln x = 1$ has exactly one solution.

From this foundation everything else follows without any leaps of faith. Since exp is the inverse of ln, the chain rule applied to $\ln(\exp x) = x$ gives

$$\exp'(x) = \exp(x),$$

so the inverse of the logarithm is its own derivative—we have recovered the calculus property as a theorem rather than asserting it. Writing $e^x := \exp(x)$ is then justified: $\exp(1) = e$ by construction, and the logarithm laws force $\exp(x) = e^x$ in the ordinary sense for rational x, which extends to all real x by continuity. The limit and series definitions reappear as standard theorems: the Taylor expansion of exp about 0 is exactly $\sum x^k/k!$, and the continuous-compounding limit

$$\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = e^x$$

follows by taking logarithms and using $\ln'(1) = 1$.

7 Summary

We have met the same number through four lenses, each illuminating a different facet:

- **Compound interest / limit:** $e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n$, the value of continuous growth at 100% per period.
- **Series:** $e = \sum_{k=0}^{\infty}\frac{1}{k!}$, the fastest practical way to compute its digits.
- **Calculus property:** e is the unique base making $\frac{d}{dx}e^x = e^x$, which is what earns it the title "natural."
- **Integral / logarithm:** e is the number for which $\int_1^e \frac{dt}{t} = 1$, the cleanest rigorous starting point.

These are not four different numbers that happen to coincide; they are four descriptions of one object, and proving their equivalence is much of what an analysis course does with e. The number that first appeared as a banker's curiosity turns out to be the hinge on which the entire theory of exponential growth, and a good deal of calculus, quietly turns.